



MANIPAL SCHOOL OF INFORMATION SCIENCES

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(A constituent unit of MAHE, Manipal)

Master of Engineering – M.E in Computer Science and Engineering

Course Plan

Course Name	:	Data Structures and Algorithms Lab
Course Code	:	ESE 5152
Academic Year	:	2026-2027
Semester	:	I
Name of the Course Coordinator	:	Jyothsna V Shetty
Name of the Program Coordinator	:	Dr. Sathyendranath Malli

Date:	08-08-2026
Signature of Program Coordinator	
with Date	

Date:	07-08-2026
Signature of Course Coordinator	
with Date	



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

Table of Contents	
1 Course Plan	5
1.1 Primary Information	5
1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping	5
1.3 Assessment Plan	6
1.4 Lesson Plan	7
1.5 References	9
1.6 Other Resources (Online, Text, Multimedia, etc.)	9
1.7 Course Timetable	10
1.8 Assessment Plan	11
1.9 Assessment Details	13
1.10 Course Articulation Matrix	13



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering – M.E in Computer Science and Engineering, programme are as follows

PEO No.	Education Objective
PEO 1	Enable to draw upon fundamental and advanced knowledge to apply analytical and computational approaches to solve technological problems in computer science.
PEO 2	Introduce state of art technologies in computer science and inculcate ethical practices to make industry-ready professionals.
PEO 3	Promote scientific and societal advancement through research and entrepreneurship.



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MANIPAL

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Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering – M.E in Computer Science and Engineering, graduates will be able to:

PO 1	An ability to independently carry out research /investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	An ability to develop software based on theoretical principles and practical knowledge.
PO 5	An ability to demonstrate knowledge of the underlying principles and evaluation methods for analysing and decision-making.



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MANIPAL

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1. Course Plan

1.1 Primary Information

Course Name	:	Data Structures and Algorithms Lab [ESE 5152]
L-T-P-C	:	0-0-3-1
Contact Hours	:	36 Hours
Pre-requisite	:	C Programming
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No. of Contact Hours	Program Outcomes (POs)	BL
CO1	Implement elementary data structures.	12	PO5	3
CO2	Design programs for sorting and searching.	12	PO4	3
CO3	Design programs for dictionary, hash tables, graphs, and trees techniques.	12	PO3	4



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1.3 Assessment Plan

Components	Lab test	Flexible assessments	End semester exam (ESE) / Makeup exam
Duration	90 minutes	Assignments are to be submitted within one week of posting	180 minutes
Weightage	0.3	0.2	0.5
Typology of questions	Apply and Analyse	Apply and Analyse	Apply and Analyse
Pattern	Answer all the Questions Maximum Marks:30	Designing the Data Structure, Problems in Lists, Stacks, Queues, Trees	Answer all the Questions: Maximum Marks:50
Schedule	As per academic calendar.	Assignment submission: October 2026	As per academic calendar.
Topics covered	Linked Lists, Stacks, Queues, Sorting and Trees	Linked Lists, Stacks, Queues, Sorting and Trees	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.



1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Linked list: defining data structure, initialization, insertion, deallocation	C01
L2	Linked list: searching, insertion in between nodes, problem solving	C01
L3	Linkedlist: deletion	C01
L4	Stack: push pop and peek in array implementation	C01
L5	Stack: push pop and peek in linked list implementation	C01,C02
L6	Queue and circular queue	C01,C02
L7	Sorting algorithms – Selection, Bubble, Insertion, Merge Sort, Quick, Heap Sort	C02
L8	Sorting algorithms – Selection, Bubble, Insertion, Merge Sort, Quick, Heap Sort	C02
L9	Trees – Binary Search Tree Traversal	C03
L10	Graph – Implementing Adjacency List,DFS	C03
L11	Hash Table	C03



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL
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L12	Assignment completion	C03
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1.5 References

1. Introduction to Algorithms – Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest. MIT Press.
2. Data Structures and Algorithms – Aho, Hopcroft and Ulmann. Pearson Publishers.

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.



MANIPAL SCHOOL OF INFORMATION SCIENCES

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1.7 Course Timetable

Semester - I						Lab:	Embedded Systems Lab		
	8-9	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Monday									
Tuesday									
Wednesday									
Thursday									
Friday									
Saturday									
							DSA Lab		



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

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1.8 Assessment Plan

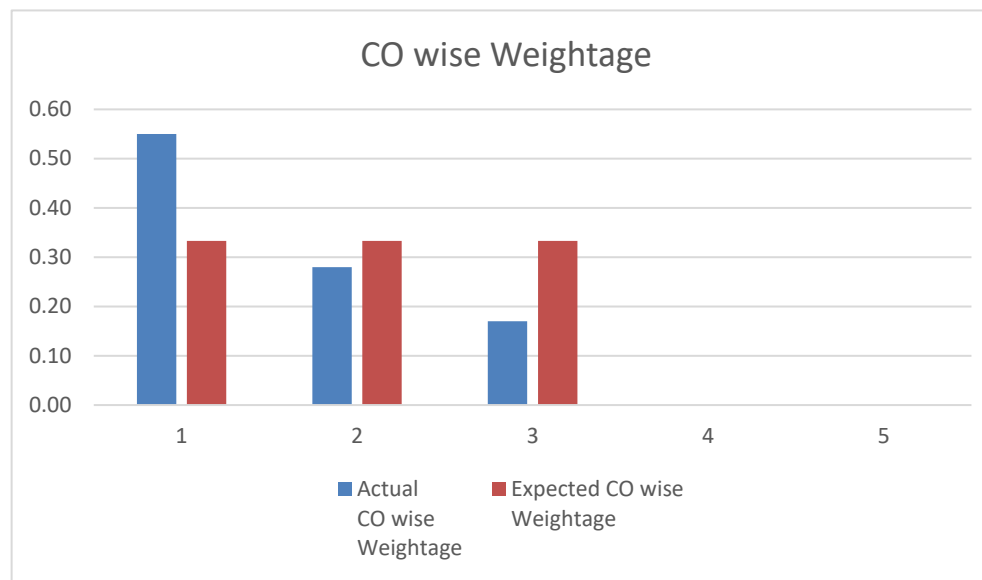
COs		Marks & Weightage				
CO No.	CO Name	Lab test marks (Max. 30)	Assignment Marks (Max. 20)	ESE Marks (Max. 50)	Actual CO wise Weightage	Expected CO wise Weightage
CO1	Implement elementary data structures.	15	10	30	0.55	0.33
CO2	Design programs for sorting and searching.	15	5	8	0.28	0.33
CO3	Design programs for dictionary, hash tables, graphs, and trees techniques.		5	12	0.17	0.33
	Marks (weightage)	0.3	0.2	0.5	1.00	1.00



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Note:

- In-semester Assessment is considered as the Internal Assessment (IA) in this course for 50 marks, which includes performances in lab participation, assignment work, lab work, lab test, quizzes etc.
- ESE for this course is conducted for a maximum of 50.
- ESE marks for a maximum of 50 and IA marks for a maximum of 50 are added for a maximum of 100 marks to decide upon the grade in this course.

Weightage for CO1 = (Lab test marks for CO1 + Assignment marks for CO1 + ESE marks for CO1) /100

$$= (10 + 5+10)/100 = 0.25$$



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

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1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

Sl. No.	Tools	Weightage	Frequency	Details of Measurement (Weightage/Rubrics/Duration, etc.)
1	Lab test	0.3	1	• Performance is measured using lab test attainment level. • Reference: question paper and answer scheme. • The lab test is assessed for a maximum of 30 marks.
2	Assignments	0.2	1	• Performance is measured using assignments attainment level. • Assignment is evaluated for a maximum of 20 marks.
3	ESE	0.5	1	• Performance is measured using ESE attainment level. • Reference: question paper and answer scheme. • ESE is assessed for a maximum of 50 marks.

1.10 Course Articulation Matrix

CO	P01	P02	P03	P04	P05
C01					Y
C02				Y	
C03			Y		
Average Articulation Level			*	*	*